

COMP3007 HCI

National Rail Website Usability Evaluation

A UX Analysis and Redesign Proposal



University of Plymouth



February 2026

Presentation Overview

1

1 minute

Software Selection & Evaluation

Understanding National Rail's purpose, evaluation methodology, participant details, and ethical documentation



6 required documents

2

1.5 minutes

Analysis & Redesign Process

Data collection methods, analysis approach, and comprehensive issue identification with prioritisation



Essential & Desirable lists

3

3.5 minutes

Redesign Solutions

Before/after examples demonstrating UX improvements to address each identified usability issue



5+ redesign proposals



Total presentation time: **6 minutes**



Strictly following assessment requirements

Section 1

Software Selection & Evaluation

Understanding the National Rail website and our evaluation methodology



Website
nationalrail.co.uk



Participants
7+ evaluators

National Rail Enquiries: Why the national rail???

Website Overview

National Rail Enquiries (nationalrail.co.uk) is the UK's official rail journey service, providing comprehensive train travel information across Great Britain.



Primary Purpose

- ✓ Plan train journeys across the UK rail network
- ✓ Check train times, routes, and platform information
- ✓ Find and compare ticket prices
- ✓ Access live service updates and disruption alerts
- ✓ Obtain travel information including accessibility

Target Users



Commuters



Leisure Travelers



Tourists



Accessibility Needs

Why This Website?

-  High daily usage with millions of journey plans
-  Complex information architecture
-  Significant potential for usability improvements
-  This site is extremely popular

Evaluation Methodology & Participants

Evaluation Approach

Conducted **usability testing** using the **think aloud protocol**, where testers verbalised their thoughts out loud to me while completing tasks.

Key Protocol Elements:

Verbalise thoughts

Express expectations

Describe confusion

7 Task Scenarios

1 Find a train journey

2 Change time/date

3 Check ticket restrictions

4 Accessibility info

5 Plan for disruption

6 Railcard

7 Compensation

Participants

7

Total Participants

These are the requirements to be eligible to participate, according to the ethics document

Plymouth University students



Family members



Friends



Age requirement: 18+



Data Collection

-  Researcher observation checklist
-  Task completion tracking
-  Time measurement per task
-  Verbal comments capture
-  Exit questionnaire if they want to withdraw

Ethical Documentation

All documents displayed for 8-10 seconds total, in specified order

1

Ethics Approval

University of Plymouth



Faculty of Science and Engineering

Research Ethics Committee Approved

2

Research Information Sheet

Participant Information



Project purpose, procedures, benefits, confidentiality

Informed consent documentation

3

Right to Withdraw

Participant Slip



Voluntary withdrawal at any time

Data destruction option included

4

Ethics Consent Form

Signed Agreement



Participant signature confirming understanding

Age verification (18+)

5

Task Scenarios

7 Evaluation Tasks



Journey planning, restrictions, accessibility,
compensation

Success criteria defined

6

Exit Questionnaire

Post-Session Feedback



10 questions on ease of use, confusion points, trust

Anonymous feedback collection

Pause video to read each document

Section 2

Analysis & Redesign Process

From raw data to actionable design decisions



Analysis Method
Affinity Mapping



Issues Identified
Essential & Desirable

Data Collection & Analysis Approach

Data Collection Methods

Researcher Observation Checklist

Tracked navigation patterns, click paths, hesitation points, error occurrences, and any questions during each task

Task Completion Tracking

Recorded completion status (completed/partially/failed) and time taken for each of the 7 tasks

Verbal Comments Capture

Documented participant quotes and think-aloud observations revealing confusion and expectations

Analysis Process

1 Affinity Mapping

Grouped similar issues across all participants to identify recurring patterns and themes

2 Frequency Analysis

Counted how many participants encountered each issue to determine prevalence

3 Connection to Interface Elements

Noted specific UI components, labels, and design decisions

4 Evidence Synthesis

Combined quantitative data (completion rates, times) with qualitative feedback (quotes, observations)

Data-to-Decision Pathway

Raw observation data → Pattern identification → Issue categorisation → Priority ranking → Design solution mapping

Identified Usability Issues

Grouped into Essential (High Priority) and Desirable (Medium Priority) categories



Essential Issues

High Priority – Critical Impact

1

Confusing Ticket Terminology

Terms 'Off-Peak', 'Anytime', 'Restriction code' unclear to **43% of participants** (P2, P3, P4)

"I don't really know what Off-Peak means" – P2

2

'Delay Repay' Term Confusion

4 participants questioned this compensation terminology

"Why isn't this just called refunds?" – P1

3

Disruption Info Not Prominent

Users expected live updates visible immediately on homepage

"I thought delays would be on the homepage" – P1

4

Broken Link. 'Mature Students' page returns 404 error

Users is presented 'Mature Students' as a clickable link but it leads to a 404 error page with no alternative guidance provided

"Wait, so how am I meant to know which railcard is for me>" – P6



Desirable Issues

Medium Priority – Enhancement

1

Visual Clutter on Results Page

Multiple competing elements create an overstimulating UI

"There's a lot on this page" – P2

2

Ticket Restriction Info is hard to find

Location and explanation of restrictions unclear

Task 3: 50% partial completion rate

3

Help Section Guidance is unclear

Journey day preparation information not clear enough

P2 & P3: "Guidance clear: No"

4

Railcard Options Presented as Flat List, rather than a Decision Based structure

Users must interpret age rules and eligibility across multiple links with no help or guidance

No grouping at all. Groups like age, student status, disability would help greatly.

Section 3

Redesign Solutions

Proposed interface improvements to address identified usability issues



Format

Before/After Screenshots



Highlighting

Changed Areas Marked

Clearer Ticket Terminology

Addressing confusion with 'Off-Peak', 'Anytime', and 'Restriction code'

✖

BEFORE

Ticket Options

Off-Peak Return

✖ Confusing

£45.60

Anytime Day Single

✖ Unclear meaning

£62.30

Restriction Code: 1A

✖ Technical jargon

View

"I don't really know what Off-Peak means" - P2

"This wording isn't great" - P4

✔

AFTER

Ticket Options

Off-Peak Return

✔ Clear

£45.60

Travel after 9:30 AM, valid on any train

Anytime Day Single

✔ Clear

£62.30

Valid all day on any train

Travel Conditions

✔ Plain English

View Details

What does this mean?

✔ Plain English explanations

✔ Contextual tooltips available

Adding this information allows for deeper understanding from the user, which leads to a more seamless interaction

'Delay Repay' to 'Compensation'

Replacing confusing industry jargon with familiar terminology

✖

BEFORE

Help & Support

£ Delay Repay

✖ Industry jargon

Claim compensation for delayed trains

Participant Feedback:

"Why isn't this just called refunds?" – P1

"Why is this called Delay Repay?" – P2

"It's not very clear" – P3

"They should just call this compensation" – P4

57% of participants confused

✔

AFTER

Help & Support

£ Compensation & Refunds

✔ Familiar terms

Get money back for delayed or cancelled trains

ℹ Delay Repay Scheme

Our compensation program for train delays of 15+ minutes

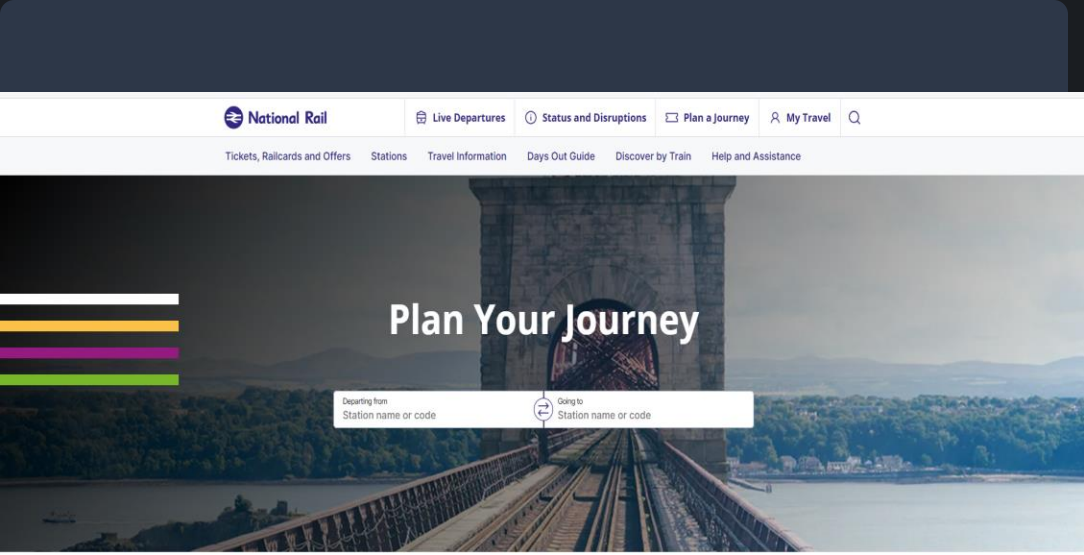
15–29 min: 25%30–59 min: 50%60+ min: 100%

✔ Clear & intuitive

Prominent Disruption Alerts

Making service updates immediately visible on the homepage

BEFORE



 Service Updates

At the top as a "Status and Disruptions button", is not clear for some users. A live status would be easier to notice

"I thought delays would be on the homepage" – P1

"It should say delays clearly at the top," – P3

AFTER

Live Service Status

View All Updates

All lines running normally

National Rail Enquiries

Your journey starts here

FROM → TO

Plan Your Journey

- ✓ Prominent banner at top of page
- ✓ Color-coded status indicators
- ✓ Real-time updates visible immediately

Simplified Search Results Layout

Reducing visual clutter and improving information hierarchy



BEFORE

Plymouth → London Paddington

 Rail Replacement Service between Plymouth and Tiverton Parkway

 All or part of this journey will be made by bus

10:05

Plymouth (PLY)

→

14:12

London Paddington (PAD)

Platform 8

Single from

£74.90

Select

4h 7m, 2 changes



- Issues:
- Multiple competing visual elements
 - Information overload per result
 - Poor visual hierarchy
 - Hard to scan quickly

"There's a lot on this page" - P2



AFTER

Plymouth → London Paddington

10:05 → 14:12

£45.60

4h 7m · 2 changes · Off-Peak

Fastest route

Select

10:15 → 13:20

£62.30

3h 5m · Direct · Anytime

Direct service

Select

- Improvements:
- ✓ Clearer visual hierarchy
 - ✓ Reduced visual noise
 - ✓ Better spacing and grouping
 - ✓ Easier to scan quickly
- ✓ Progressive disclosure for details

Improved Ticket Restriction Information

Making travel conditions understandable without technical jargon

✖

BEFORE

Ticket Details

Off-Peak Return

£45.60

Restriction Code: 1A

Technical code without explanation

View

What does this mean?

Passengers must click through to understand restrictions, leading to confusion and potential ticket misuse.

"What does restriction code mean?" - P1

"I don't know the difference" (Off-Peak/Anytime) - P3

✔

AFTER

Ticket Details

Off-Peak Return

£45.60

ⓘ

Travel Conditions

✔

Valid after 9:30 AM on weekdays

✔

Any time on weekends & bank holidays

✔

Return within 1 month

View full terms & conditions

💡

Quick Summary

Perfect for flexible travel after morning rush hour. Save money by avoiding peak times!

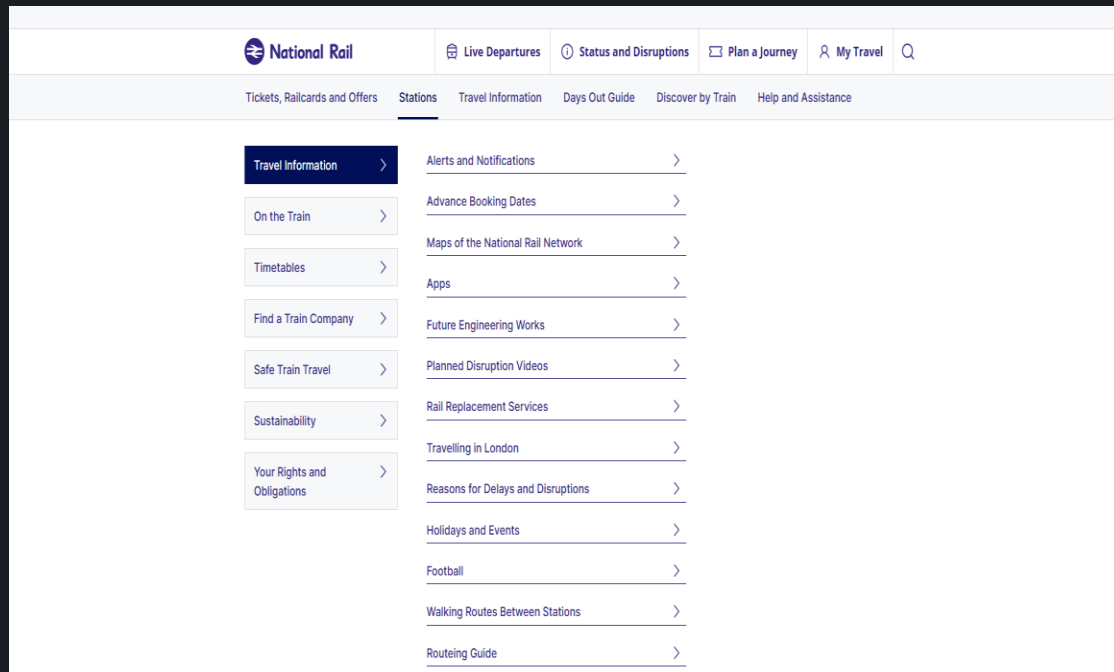
✔ Plain English explanations

✔ Visual icons for quick understanding

✔ Expandable for full details

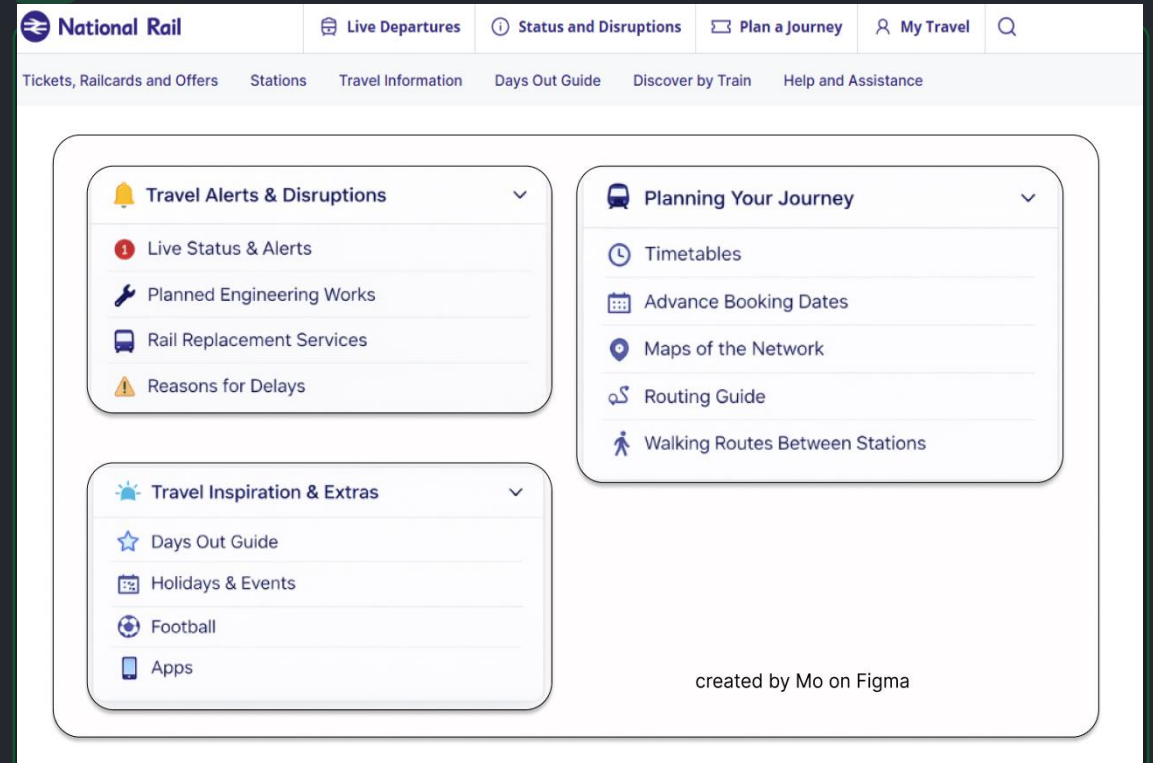
Improved menu

BEFORE



Cluttered and messy, not easy on the eyes

AFTER



Easier to view, easier to read. Icons allow for some colour to be added too.

Summary & Impact

✓ Issues Addressed

- ✓ **Clearer Terminology**
Replaced jargon with plain English explanations
- ✓ **Prominent Alerts**
Made disruption information immediately visible
- ✓ **Cleaner Layouts**
Reduced visual clutter and improved hierarchy
- ✓ **Better Guidance**
Made ticket conditions understandable

💡 HCI Principles Applied


Visibility of System Status


Match System to Real World


Aesthetic & Minimalist Design


Help & Documentation

📈 Expected Outcomes

- ↑ Reduced cognitive load for users
- ↑ Faster task completion times
- ↑ Improved user satisfaction
- ↑ Fewer errors and confusion points

All proposed changes are grounded in participant feedback and established HCI principles, ensuring evidence-based design decisions that address real user needs.